

Matrix Distribution:

Using $p = n^2$ processes, Rank 0 does:

- Generate full matrices A and B using a fillArray() function.
- Send **one element from A and one from B** to each process using MPI_Send.
- Each Process receives

one element of matrix A

. one element of matrix B

Compute a single result element $C[i][j]$

Exchange & Computation:

- **A Broadcast:** Each process row broadcasts A values using MPI_Bcast. Root shifts each iteration via $(\text{row} + k) \% n$.
 - **B Shift:** B values are shifted up with wraparound using MPI_Sendrecv_replace for simultaneous send/receive.
 - **Computation:** Each process multiplies and accumulates results over n iterations.
 - **Result Gathering:** Final values are sent to Rank 0 with MPI_Send, which reconstructs the full matrix C.
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MPI Communication with Reason:

Purpose	Type	Mode	MPI Function(s)	Reason
Sending A and B values	Point-to-point	Blocking	MPI_Send, MPI_Recv	Send single elements from Rank 0 to all processes
Row-wise broadcast of A	Collective	Blocking	MPI_Bcast	Efficiently broadcast A value within a row of processes
Circular shift of B (vertical)	Point-to-point	Blocking	MPI_Sendrecv_replace	Simultaneous send/recv for rotating B values between rows
Gathering C values	Point-to-point	Blocking	MPI_Send, MPI_Recv	Gather computed results back at Rank 0 to form the final matrix